Pandas

Study Plan & Resources

Everything you need to learn Pandas — from the basics to real EDA and ML workflows.

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Study Topics

Part 1

Core Pandas Topics

The fundamental concepts every student must know before applying Pandas to real data.

Topic	What to Know
Data Structures	Series and DataFrame — how data is stored, indexed, and accessed in Pandas.
Loading Data	Reading CSV, Excel, and JSON files. Understanding index and column assignment on load.
Indexing & Selection	loc, iloc, boolean masking, and conditional filtering to slice your data precisely.
Data Types	Checking and converting dtypes — knowing when a column is the wrong type and how to fix it.
Handling Missing Data	Detecting nulls with isnull(), and deciding when to drop, fill, or flag them.
Sorting & Ranking	sort_values(), sort_index(), and rank() to order and compare data.
Grouping & Aggregation	groupby() with agg(), mean(), count(), sum() — the core of any EDA workflow.
Merging & Joining	merge(), join(), and concat() — combining datasets horizontally and vertically.
String Operations	str accessor methods for cleaning and transforming text columns.
Apply & Map	Applying custom functions across rows or columns with apply(), map(), and applymap().

Part 2

Important Use Cases

How the topics above connect to real EDA and machine learning pipelines.

Once you know the tools, you need to know when and why to use them. These are the key workflows that matter in practice:

Exploratory Data Analysis (EDA)

Using describe(), value_counts(), groupby(), and correlation matrices to understand a dataset before modeling.

Data Cleaning for ML

Handling nulls, fixing dtypes, encoding categoricals, and removing or capping outliers so your data is model-ready.

Feature Engineering

Creating new columns from existing ones — ratios, bins, date parts, and interaction terms — to give models more signal.

Aggregation & Reporting

Summarizing data by group to answer business questions: top performers, category breakdowns, time trends.

Merging Datasets

Combining multiple tables correctly — understanding inner vs outer joins and avoiding duplicate rows or lost records.

Preprocessing Pipelines

Structuring your cleaning and transformation steps in a logical, reproducible order before passing data to sklearn.

Part 3

Resources

Curated materials to help you learn and go deeper — start with the team's own content first.

From the AI Sub-Team

VIDEO **Pandas**

Our team's own Material — start here before going anywhere else.

[Session]

SLIDES **Our Pandas Slides**

Our team's own Slides..

[Slides]

Playlists & External Guides

PLAYLIST **Playlist 1**

[Complete Python Pandas Data Science Tutorial!]

PLAYLIST **Playlist 2**

[Learn Pandas in 1 hour! 🐼]

Good luck! — Marwan Abbas | Vice Head of AI Sub-Team | IEEE SSCS